

Quantum Key Distribution: Ekert E91 Protocol Derivations

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Contents

1	Abstract	1
2	Introduction	2
3	Ekert E91 Protocol	2
4	References	9

1 Abstract

The Ekert protocol proposed by Artur Ekert in 1991 is a method for quantum key distribution. The protocol is based on the principle of quantum entanglement and a clever use of the identities prescribed by Bell's test to entangled qubits. We show below the derivations to detect an eavesdropper, Eve, when Alice and Bob are trying to exchange keys for secure communication.

2 Introduction

3 Ekert E91 Protocol

- The three bases for the three directions of measurement, $0^0, 120^0, 240^0$, are given by

$$0^0 = (|\uparrow\rangle, |\downarrow\rangle) = \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = a \quad (1)$$

$$120^0 = (|\searrow\rangle, |\swarrow\rangle) = \left(\begin{bmatrix} \frac{1}{2} \\ -\frac{\sqrt{3}}{2} \end{bmatrix}, \begin{bmatrix} \frac{\sqrt{3}}{2} \\ \frac{1}{2} \end{bmatrix} \right) = b \quad (2)$$

$$240^0 = (|\swarrow\rangle, |\searrow\rangle) = \left(\begin{bmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{bmatrix}, \begin{bmatrix} \frac{\sqrt{3}}{2} \\ -\frac{1}{2} \end{bmatrix} \right) = c \quad (3)$$

Consider two qubits which are in the entangled spin state,

$$\frac{1}{\sqrt{2}} |\uparrow\rangle |\uparrow\rangle + \frac{1}{\sqrt{2}} |\downarrow\rangle |\downarrow\rangle \quad (4)$$

Alice receives one qubit from the entangled pair and Bob the other. They continue to receive a stream of these entangled qubits.

Note that we use the following short form, $|\uparrow\rangle |\uparrow\rangle$, for the tensor product, $|\uparrow\rangle \otimes |\uparrow\rangle$, of two state vectors,

$$|\uparrow\rangle |\uparrow\rangle \equiv |\uparrow\rangle \otimes |\uparrow\rangle \quad (5)$$

We know that $3n$ qubits have been sent. Alice and Bob make measurements on these qubits by randomly using one of the three bases given in Equations (1;2;3). After they make the measurements they reveal the sequence of bases that they chose. They are only revealing the basis they chose, not the result of the measurements. They will agree on approximately n of them. We assume that n is a fairly large number. In each place they have chosen the same basis they will have made the same measurement. They will both either have 0, or both have 1. They will get either a 0 or a 1 with equal probability but they will both get exactly the same answer. This string of n 0s and 1s will be their key if Eve is not listening in.

They test for whether Eve is eavesdropping or not. If Eve is listening in, she will have to make measurements

and whenever a measurement is done, the entangled states become unentangled. They check the other $2n$ strings of 0 and 1 that come from the times when they chose different bases. Depending on whether Eve is eavesdropping or not gives different values for the proportion of qubits that they agree on. We will prove the below two properties.

1. **If the states are entangled when Alice and Bob make their measurements, then from the Bell Inequality they know that in each place they should agree only 1/4 of the time (Equation 62).**
2. **However if Eve is measuring one of the qubits the proportion of times they agree changes. The proportion of times they agree will increase to 3/8.**

- We write each of the three bases in terms of other two bases as follows:

$$|\uparrow\rangle = \alpha |\searrow\rangle + \beta |\swarrow\rangle \quad (6)$$

$$|\downarrow\rangle = \gamma |\searrow\rangle + \delta |\swarrow\rangle \quad (7)$$

We can multiply both sides of the above equations by $\langle\searrow|$ and $\langle\swarrow|$ to get,

$$\langle\searrow|\uparrow\rangle = \alpha \langle\searrow|\searrow\rangle + \beta \langle\searrow|\swarrow\rangle \quad (8)$$

$$\langle\swarrow|\uparrow\rangle = \alpha \langle\swarrow|\searrow\rangle + \beta \langle\swarrow|\swarrow\rangle \quad (9)$$

$$\langle\searrow|\downarrow\rangle = \gamma \langle\searrow|\searrow\rangle + \delta \langle\searrow|\swarrow\rangle \quad (10)$$

$$\langle\swarrow|\downarrow\rangle = \gamma \langle\swarrow|\searrow\rangle + \delta \langle\swarrow|\swarrow\rangle \quad (11)$$

Noting that, $\langle\searrow|\searrow\rangle = \langle\swarrow|\swarrow\rangle = 1$ and $\langle\searrow|\swarrow\rangle = \langle\swarrow|\searrow\rangle = 0$ gives,

$$\alpha = \langle\searrow|\uparrow\rangle = \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{2} \quad (12)$$

$$\beta = \langle\swarrow|\uparrow\rangle = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{\sqrt{3}}{2} \quad (13)$$

$$\gamma = \langle \searrow | \downarrow \rangle = \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = -\frac{\sqrt{3}}{2} \quad (14)$$

$$\delta = \langle \swarrow | \downarrow \rangle = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{2} \quad (15)$$

The calculation of the probability amplitudes can be simplified and written as,

$$\begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{bmatrix} \quad (16)$$

$$\begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -\frac{\sqrt{3}}{2} \\ \frac{1}{2} \end{bmatrix} \quad (17)$$

Using these values gives Equations (18; 19),

$$|\uparrow\rangle = \frac{1}{2} |\searrow\rangle + \frac{\sqrt{3}}{2} |\swarrow\rangle \quad (18)$$

$$|\downarrow\rangle = -\frac{\sqrt{3}}{2} |\searrow\rangle + \frac{1}{2} |\swarrow\rangle \quad (19)$$

- Similarly we need to find the probability amplitudes below (24; 25),

$$|\searrow\rangle = \alpha |\uparrow\rangle + \beta |\downarrow\rangle \quad (20)$$

$$|\swarrow\rangle = \gamma |\uparrow\rangle + \delta |\downarrow\rangle \quad (21)$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ -\frac{\sqrt{3}}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ -\frac{\sqrt{3}}{2} \end{bmatrix} \quad (22)$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{3}}{2} \\ \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{2} \\ \frac{1}{2} \end{bmatrix} \quad (23)$$

$$|\searrow\rangle = \frac{1}{2} |\uparrow\rangle - \frac{\sqrt{3}}{2} |\downarrow\rangle \quad (24)$$

$$|\swarrow\rangle = \frac{\sqrt{3}}{2} |\uparrow\rangle + \frac{1}{2} |\downarrow\rangle \quad (25)$$

- We next consider (30; 31),

$$|\uparrow\rangle = \alpha |\swarrow\rangle + \beta |\nearrow\rangle \quad (26)$$

$$|\downarrow\rangle = \gamma |\swarrow\rangle + \delta |\nearrow\rangle \quad (27)$$

$$\begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{bmatrix} \quad (28)$$

$$\begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{2} \\ -\frac{1}{2} \end{bmatrix} \quad (29)$$

$$|\uparrow\rangle = \frac{1}{2} |\swarrow\rangle + \frac{\sqrt{3}}{2} |\nearrow\rangle \quad (30)$$

$$|\downarrow\rangle = \frac{\sqrt{3}}{2} |\swarrow\rangle - \frac{1}{2} |\nearrow\rangle \quad (31)$$

- We next consider (36; 37),

$$|\swarrow\rangle = \alpha |\uparrow\rangle + \beta |\downarrow\rangle \quad (32)$$

$$|\nearrow\rangle = \gamma |\uparrow\rangle + \delta |\downarrow\rangle \quad (33)$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{bmatrix} \quad (34)$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{\sqrt{3}}{2} \\ -\frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{2} \\ -\frac{1}{2} \end{bmatrix} \quad (35)$$

$$|\swarrow\rangle = \frac{1}{2} |\uparrow\rangle + \frac{\sqrt{3}}{2} |\downarrow\rangle \quad (36)$$

$$|\nearrow\rangle = \frac{\sqrt{3}}{2} |\uparrow\rangle - \frac{1}{2} |\downarrow\rangle \quad (37)$$

- We next consider (42; 43),

$$|\searrow\rangle = \alpha |\swarrow\rangle + \beta |\nearrow\rangle \quad (38)$$

$$|\nwarrow\rangle = \gamma |\swarrow\rangle + \delta |\nearrow\rangle \quad (39)$$

$$\begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ -\frac{\sqrt{3}}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{4} - \frac{3}{4} \\ \frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{4} \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{bmatrix} \quad (40)$$

$$\begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} \frac{\sqrt{3}}{2} \\ \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{4} \\ \frac{3}{4} - \frac{1}{4} \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{2} \\ \frac{1}{2} \end{bmatrix} \quad (41)$$

$$|\searrow\rangle = -\frac{1}{2}|\swarrow\rangle + \frac{\sqrt{3}}{2}|\nearrow\rangle \quad (42)$$

$$|\nwarrow\rangle = \frac{\sqrt{3}}{2}|\swarrow\rangle + \frac{1}{2}|\nearrow\rangle \quad (43)$$

- We next consider (48; 49),

$$|\swarrow\rangle = \alpha|\searrow\rangle + \beta|\nwarrow\rangle \quad (44)$$

$$|\nearrow\rangle = \gamma|\searrow\rangle + \delta|\nwarrow\rangle \quad (45)$$

$$\begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{4} - \frac{3}{4} \\ \frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{4} \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{bmatrix} \quad (46)$$

$$\begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} \frac{\sqrt{3}}{2} \\ -\frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{4} \\ \frac{3}{4} - \frac{1}{4} \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{2} \\ \frac{1}{2} \end{bmatrix} \quad (47)$$

$$|\swarrow\rangle = -\frac{1}{2}|\searrow\rangle + \frac{\sqrt{3}}{2}|\nwarrow\rangle \quad (48)$$

$$|\nearrow\rangle = \frac{\sqrt{3}}{2}|\searrow\rangle + \frac{1}{2}|\nwarrow\rangle \quad (49)$$

- These set of six pairs of equations are summarized below,

$$|\uparrow\rangle = \frac{1}{2}|\searrow\rangle + \frac{\sqrt{3}}{2}|\nwarrow\rangle \quad (50)$$

$$|\downarrow\rangle = -\frac{\sqrt{3}}{2}|\searrow\rangle + \frac{1}{2}|\nwarrow\rangle \quad (51)$$

$$|\searrow\rangle = \frac{1}{2}|\uparrow\rangle - \frac{\sqrt{3}}{2}|\downarrow\rangle \quad (52)$$

$$|\nwarrow\rangle = \frac{\sqrt{3}}{2}|\uparrow\rangle + \frac{1}{2}|\downarrow\rangle \quad (53)$$

$$|\uparrow\rangle = \frac{1}{2}|\swarrow\rangle + \frac{\sqrt{3}}{2}|\nearrow\rangle \quad (54)$$

$$|\downarrow\rangle = \frac{\sqrt{3}}{2}|\swarrow\rangle - \frac{1}{2}|\nearrow\rangle \quad (55)$$

$$|\swarrow\rangle = \frac{1}{2}|\uparrow\rangle + \frac{\sqrt{3}}{2}|\downarrow\rangle \quad (56)$$

$$|\nearrow\rangle = \frac{\sqrt{3}}{2} |\uparrow\rangle - \frac{1}{2} |\downarrow\rangle \quad (57)$$

$$|\searrow\rangle = -\frac{1}{2} |\swarrow\rangle + \frac{\sqrt{3}}{2} |\nearrow\rangle \quad (58)$$

$$|\nwarrow\rangle = \frac{\sqrt{3}}{2} |\swarrow\rangle + \frac{1}{2} |\nearrow\rangle \quad (59)$$

$$|\swarrow\rangle = -\frac{1}{2} |\searrow\rangle + \frac{\sqrt{3}}{2} |\nwarrow\rangle \quad (60)$$

$$|\nearrow\rangle = \frac{\sqrt{3}}{2} |\searrow\rangle + \frac{1}{2} |\nwarrow\rangle \quad (61)$$

- Using Equation (4) we know that there is a probability of 1/2 of getting either 0 or 1 if the pair of qubits are entangled. This is because there is an equal probability of getting 0 or 1 as seen from Equation (4). Whoever measures first, they have a 1/2 probability of getting 0 or 1. This is because the entangled qubits can be expressed using the basis chosen by the person who measures first and the probability of getting 0 or 1 is 1/2 (See Proof on Page 73 and 74 of Chapter 5, Bell's Inequality in Quantum Computing by Chris Bernhardt).

We also know that Alice and Bob are measuring using different bases since we are only comparing those qubits that were measured using different bases. After the first measurement is made, the qubits are no longer entangled and they can be expressed using the basis chosen by the person who makes the second measurement. When they measure with different bases, the probability of getting 0 or 1 is 1/4. We can see this from the set of Equations (18; 19; 24; 25; 30;31;36;37;42;43;48;49) and by looking at the coefficients of the vector that represents 0 or 1. If they measure at the same time, we can still consider the qubit to be expressed in either of the bases chosen by Alice or Bob and a similar result will be obtained.

When Eve is not eaves dropping and when Alice or Bob make a measurement the qubits are entangled. The first measurement puts both the qubits in the same state., either 0 or 1 with equal probability of 1/2. Hence the probability that Alice and Bob will get the same qubits, or that they agree, when the qubits are entangled when either of them makes a measurement is given by,

$$\left(\frac{1}{2}\right) \left(\frac{1}{4}\right) + \left(\frac{1}{2}\right) \left(\frac{1}{4}\right) = \frac{1}{4} \quad (62)$$

- A key question is specifically from the last section (The Ekert Protocol for Quantum Key Distribution) from Chapter 5 (Bell's Inequality) of the book Quantum Computing for Everyone by Chris Bernhardt. Why if Eve, an eavesdropper, is measuring one of the qubits the proportion of times Alice and Bob agree changes. The proportion of times they agree will increase to 3/8. Below we clarify how we arrive at this number 3/8.
- We assume Eve, along with Alice and Bob, picks her basis at random. We are given that Alice and Bob chose

different bases. So there is a $\frac{2}{3}$ chance that Eve picks the same basis as either Alice or Bob, and a $\frac{1}{3}$ chance that she picks the remaining basis.

If Eve picks the same one as either Alice or Bob, the odds that Alice and Bob agree remain at $\frac{1}{4}$. This is the result of a calculation similar to Equation (62).

But what happens if Eve picks the remaining basis? She does her measurement, which disentangles the qubits, but gives them the same value. Alice now does her measurement. She has a $\frac{3}{4}$ chance of getting a different answer from Eve and a $\frac{1}{4}$ chance of getting the same answer as Eve. Likewise for Bob. So the probability that Alice and Bob get the same result is,

$$\left(\frac{3}{4}\right)\left(\frac{3}{4}\right) + \left(\frac{1}{4}\right)\left(\frac{1}{4}\right) = \frac{5}{8} \quad (63)$$

So all three have equal probability, the result is,

$$\frac{2}{3}\left(\frac{1}{4}\right) + \frac{1}{3}\left(\frac{5}{8}\right) = \frac{3}{8} \quad (64)$$

- Alternately, after Eve has made a measurement the qubits will be in state $|0\rangle$ or $|1\rangle$ (0 or 1) depending on the basis chosen by Eve. We denote the three basis by a, b, c as shown above in Equations (1; 2; 3). Suppose that Eve has measured using basis a . Then Alice and Bob can have the following choices of bases and the corresponding probability of agreement or of both of them getting the same result or getting 0 or 1.

aa	ab	ac	ba	bb	bc	ca	cb	cc
1	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{5}{8}$	$\frac{5}{8}$	$\frac{1}{4}$	$\frac{5}{8}$	$\frac{5}{8}$

(65)

If Alice and Bob use the same basis as Eve, they will agree on the result 100% or with probability 1. If one of them measures using the same basis as Eve, then the probability of agreement will be $\frac{1}{4}$. This is the result of a calculation similar to Equation (62). If they both measure with a basis other than the one chosen by Eve, they both have a $\frac{1}{4}$ probability of getting 0 or a $\frac{3}{4}$ probability of getting 1. Hence, the probability of agreement is given by,

$$\left(\frac{1}{4}\right)\left(\frac{1}{4}\right) + \left(\frac{3}{4}\right)\left(\frac{3}{4}\right) = \frac{5}{8} \quad (66)$$

We just need to consider the measurements where Alice and Bob are using different bases. This gives,

ab	ac	ba	bc	ca	cb
$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{5}{8}$	$\frac{1}{4}$	$\frac{5}{8}$

(67)

Each of these six choice of basis by Alice and Bob is equally likely or has probability $\frac{1}{6}$ each. Hence the probability they will agree is given by,

$$\left(\frac{1}{6}\right)\left[4\left(\frac{1}{4}\right) + 2\left(\frac{5}{8}\right)\right] = \frac{3}{8} \quad (68)$$

- Since we have three directions of measurement there is a $1/3$ probability of Eve choosing either direction of measurement. And the same probability of agreement of $3/8$ is obtained. for each direction. Hence, the result is $3/8$ as shown below,

$$\left(\frac{1}{3}\right) \left(\frac{3}{8}\right) + \left(\frac{1}{3}\right) \left(\frac{3}{8}\right) + \left(\frac{1}{3}\right) \left(\frac{3}{8}\right) = \frac{3}{8} \quad (69)$$

4 References

- Bohr, N. (1935). Can quantum-mechanical description of physical reality be considered complete?. Physical review, 48(8), 696.
- Bohm, D. (1952). A suggested interpretation of the quantum theory in terms of "hidden" variables. I. Physical review, 85(2), 166.
- Bohm, D. (1952). A Suggested Interpretation of the Quantum Theory in Terms of "Hidden" Variables. II. Physical Review, 85(2), 180.
- Bell, J. S. (1964). On the Einstein Podolsky Rosen paradox. Physics Physique Fizika, 1(3), 195.
- Bernhardt, C. (2019). Quantum computing for everyone. Mit Press.
- DiVincenzo, D. P. (1995). Quantum computation. Science, 270(5234), 255-261.
- Einstein, A., Podolsky, B., & Rosen, N. (1935). Can quantum-mechanical description of physical reality be considered complete?. Physical review, 47(10), 777.
- Ekert, A. K. (1991). Quantum cryptography based on Bell's theorem. Physical Review Letters, 67(6), 661-663.
- Freedman, S. J., & Clauser, J. F. (1972). Experimental test of local hidden-variable theories. Physical Review Letters, 28(14), 938.
- Hensen, B., Bernien, H., Dréau, A. E., Reiserer, A., Kalb, N., Blok, M. S., ... & Hanson, R. (2015). Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres. Nature, 526(7575), 682-686.
- Ilic, N. (2007). The ekert protocol. Journal of Phy334, 1, 22.
- Kimble, H. J. (2008). The quantum internet. Nature, 453(7198), 1023-1030.
- Maccone, L. (2013). A simple proof of Bell's inequality. American Journal of Physics, 81(11), 854-859.

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